Gas Laws

41. The gas equation $\frac{PV}{T}$ = constant is true for a constant mass of an ideal gas undergoing

(a) Isothermal change

(b) Adiabatic change

(c) Isobaric change

(d) Any type of change

42. That gas cannot be liquified

(a) Which obeys Vander Waal's equation

(b) Which obeys gas equation at every temperature and pressure

(c) The molecules of which are having potential energy

(d) Which is a inert gas

43. At NTP the mass of one litre of air is 1.293 *gm*. The value of specific gas constant will be

(a) 0.29 *J/K-gm* (b) 4.2 *J/K-gm*

(c) 8.3 *J/K-gm* (d) 16.5 *J/Kgm*

44. The volume of a gas at 21°*C* temperature and 768 *mm* pressure is 1 *litre*. If the density of the gas is 1.2 *gm/litre* at NTP, then its mass will be

(a) 4 *gm* (b) 4.21 *gm*

(c) 1.13 *gm* (d) 10 *gm*

45. For one gram mol of a gas, the value of R in the equation *PV* = *RT* is nearly

(a) 2 *cal/K* (b) 10 *cal/K*

(c) 0.2 *cal/K* (d) 200 *cal/K*

46. The number of molecules in a gas at pressure 1.64×10^{-3} atmospheres and temperature 200 *K* having the volume 1 *cc* are

(a) 6.02×10^{16} (b) 2.63×10^{16}

(c) 3.01×10^{19} (d) 12.04×10^{19}

47. The pressure *P*, volume *V* and temperature *T* of a gas in the jar *A* and the other gas in the jar *B* at pressure 2*P*, volume *V*/4 and temperature 2*T*, then the ratio of the number of molecules in the jar *A* and *B* will be

(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) 4 : 1

48. We write the relation for Boyle's law in the form *PV* = *C* when the temperature remains constant. In

this relation, the magnitude of C depends upon

(a) The nature of the gas used in the experiment

(b) The magnitude of g in the laboratory

(c) The atmospheric pressure

(d) The quantity of the gas enclosed

49. If a given mass of gas occupies a volume of 10 *cc* at 1 atmospheric pressure and temperature of 100°*C*(373.15 *K*). What will be its volume at 4 atmospheric pressure; the temperature being the same

(a) 100 *cc* (b) 400 *cc*

(c) 2.5 *cc* (d) 104 *cc*

50. A sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T, the mass of each molecule is m. The expression for the density of gas is (k = Boltzmann's constant)

(a) mkT (b) P/kT

(c) P/kTV (d) Pm/Kt

51. The product of the pressure and volume of an ideal gas is

(a) A constant

(b) Approx. equal to the universal gas constant

(c) Directly proportional to its temperature

(d) Inversely proportional to its temperature

52. A balloon contains $500m^3$ of helium at 27°*C* and 1 atmosphere pressure. The volume of the helium at – 3°*C* temperature and 0.5 atmosphere pressure will be

(a) $500m^3$ (b) $700m^3$

(c) $900m^3$ (d) $1000m^3$

53. A vessel contains 1 mole of O_2 gas (molar mass 32) at a temperature T. The pressure of the gas is P. An identical vessel containing one mole of *He* gas (molar mass 4) at temperature $2T$ has a pressure of

(a) $P/8$ (b) P

(c) $2P$ (d) $8P$

54. 1 mole of gas occupies a volume of 100 *ml* at 50 *mm* pressure. What is the volume occupied by two moles of gas at 100 *mm* pressure and at same temperature

(a) 50 *ml* (b) 100 *ml*

(c) 200 *ml* (d) 500 *ml*

55. For ideal gas, which statement is not true

(a) It obeys Boyle's law

(b) It follows $PV = RT$

(c) Internal energy depends on temperature only

(d) It follows Vander-Waal's equation

56. A given mass of a gas is allowed to expand freely until its volume becomes double. If C_b and C_a are the velocities of sound in this gas before and after expansion respectively, then C_a is equal to

(a) $2C_b$ (b) $\sqrt{2}C_b$

(c) C_b (d) $\frac{1}{\sqrt{2}}C_b$

57. Kinetic theory of gases was put forward by

(a) Einstein (b) Newton

(c) Maxwell (d) Raman

58. One *litre* of Helium gas at a pressure 76 *cm* of *Hg* and temperature 27° is heated till its pressure and volume are doubled. The final temperature attained by the gas is

(a) 927°*C* (b) 900°*C*

(c) 627°*C* (d) 327°*C*

59. A flask is filled with 13 *gm* of an ideal gas at 27°*C* and its temperature is raised to 52°*C*. The mass of the gas that has to be released to maintain the temperature of the gas in the flask at 52°*C* and the pressure remaining the same is

(a) 2.5 *g* (b) 2.0 *g*

(c) 1.5 *g* (d) 1.0 *g*

60. The molecular weight of a gas is 44. The volume occupied by 2.2 *g* of this gas at $0°C$ and 2 *atm.* pressure will be

(a) 0.56 *litre* (b) 1.2 *litres*

(c) 2.4 *litres* (d) 5.6 *litres*

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